

## APPENDIX

This appendix provides the model and numerical details supporting the experiments in the main paper. Section ?? defines the standalone RMS inverter, cascaded M-section line, EMT grid-following inverter, and synchronous-machine EMT benchmark models. Section E describes the construction and parameterization of the IEEE 9-, 14-, 39-, 57-, and 118-bus networked DAE benchmarks. Section F summarizes the benchmark-specific physical parameters, controller gains, initialization settings, and internal RAPTOR numerical parameters required for reproducibility. Finally, Section L reports comparisons with alternative temporal representations and classical integration methods.

### A. Inverter

1) *Model Description:* The dynamics of the grid-following converter are captured by the eleven-dimensional state vector

$$\mathbf{x} = [\xi_{\text{pll}}, \theta_{\text{pll}}, i_d, i_q, \xi_{i_d}, \xi_{i_q}, \xi_P, \xi_Q, P_{\text{filt}}, Q_{\text{filt}}, V_{\text{filt}}]^\top$$

the seven-dimensional input vector

$$\mathbf{u} = [P_{\text{ref}}, Q_{\text{ref}}, V_{\text{ref}}, \omega_{\text{ref}}, v_{ga}, v_{gb}, v_{dc}]^\top$$

and the fourteen-dimensional parameter vector

$$\mathbf{p} = [k_{p_{dp}}, k_{i_{dp}}, k_{p_{dq}}, k_{i_{dq}}, k_{p_{id}}, k_{i_{id}}, k_{p_{iq}}, k_{i_{iq}}, k_{p_{\text{pll}}}, k_{i_{\text{pll}}}, \omega, L, R, \omega_{\text{filt}}]^\top. \quad (1)$$

Measured phase-to-ground voltages  $v_{ga}, v_{gb}$  are first transformed into the synchronous  $dq$  frame by

$$v_{gd} = v_{ga} \cos \theta_{\text{pll}} + v_{gb} \sin \theta_{\text{pll}}, \quad (2)$$

$$v_{gq} = -v_{ga} \sin \theta_{\text{pll}} + v_{gb} \cos \theta_{\text{pll}}. \quad (3)$$

These signals drive a phase-locked loop (PLL) whose PI controller uses gains  $k_{p_{\text{pll}}}, k_{i_{\text{pll}}}$  to process the quadrature component  $v_{gq}$ . The resulting frequency correction  $\omega_{\text{pll}} = k_{p_{\text{pll}}} v_{gq} + k_{i_{\text{pll}}} \xi_{\text{pll}}$  is added to the reference  $\omega_{\text{ref}}$  to yield the total angular speed  $\omega = \omega_{\text{ref}} + \omega_{\text{pll}}$ . The PLL integrator and angle update are

$$\dot{\xi}_{\text{pll}} = v_{gq}, \quad \dot{\theta}_{\text{pll}} = \omega.$$

The outer control loop regulates active and reactive power. Power errors are defined by

$$dP = P_{\text{ref}} - P_{\text{filt}}, \quad dQ = -Q_{\text{ref}} + Q_{\text{filt}},$$

and a secondary PI stage with gains  $k_{p_{dp}}, k_{i_{dp}}$  and  $k_{p_{dq}}, k_{i_{dq}}$  generates current references

$$i_{d,\text{ref}} = k_{p_{dp}} dP + k_{i_{dp}} \xi_P, \quad (4)$$

$$i_{q,\text{ref}} = k_{p_{dq}} dQ + k_{i_{dq}} \xi_Q, \quad (5)$$

while the integrator states obey

$$\dot{\xi}_P = dP, \quad \dot{\xi}_Q = dQ.$$

In the inner current control loop, the  $d$ - and  $q$ -axis current errors are fed to PI regulators with gains  $k_{p_{id}}, k_{i_{id}}$  and  $k_{p_{iq}}, k_{i_{iq}}$ . These produce converter voltage commands

$$v_{md} = k_{p_{id}}(i_{d,\text{ref}} - i_d) + k_{i_{id}} \xi_{i_d} - \frac{\omega}{\omega_{\text{ref}}} L i_q, \quad (6)$$

$$v_{mq} = k_{p_{iq}}(i_{q,\text{ref}} - i_q) + k_{i_{iq}} \xi_{i_q} + \frac{\omega}{\omega_{\text{ref}}} L i_d, \quad (7)$$

with integrator dynamics

$$\dot{\xi}_{i_d} = i_{d,\text{ref}} - i_d, \quad \dot{\xi}_{i_q} = i_{q,\text{ref}} - i_q.$$

The electrical network seen by the converter is modeled as a series  $RL$  impedance in the synchronous frame. The current dynamics are

$$\dot{i}_d = -\frac{R}{L} i_d + \omega i_q + \frac{1}{L} (v_{md} - v_{gd}), \quad (8)$$

$$\dot{i}_q = -\frac{R}{L} i_q - \omega i_d + \frac{1}{L} (v_{mq} - v_{gq}). \quad (9)$$

Finally, the converter's modulated voltage magnitude  $v_m = \sqrt{v_{md}^2 + v_{mq}^2}$  and current magnitude  $i_m = \sqrt{i_d^2 + i_q^2}$  yield instantaneous power

$$P = v_{md} i_d + v_{mq} i_q, \quad Q = v_{mq} i_d - v_{md} i_q.$$

These signals are filtered by first-order dynamics

$$\dot{P}_{\text{filt}} = (P - P_{\text{filt}}) \omega_{\text{filt}}, \quad \dot{Q}_{\text{filt}} = (Q - Q_{\text{filt}}) \omega_{\text{filt}}, \quad \dot{V}_{\text{filt}} = (v_m - V_{\text{filt}}) \omega_{\text{filt}}.$$

Together, these continuous dynamics describe the nested PLL, power, and current control loops, the converter's electrical behavior, and the measurement filtering in a unified state-space framework.

### B. M-section Line Model

The M-section line model is a mixture of the T-section and  $\pi$ -section models, which allows most of the potential and magnetic energy of the cable to be stored within the cable section while the cable terminal voltages have faster dynamics. The model is shown in Fig. 1.

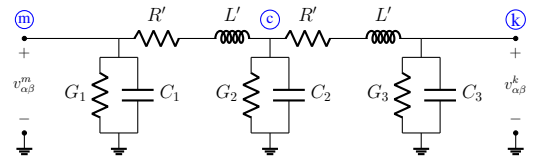


Fig. 1:  $m$ -section line model.

The dynamics of the cascaded three-section model are captured by the ten-dimensional state vector

$$\mathbf{x} = [v_a^m, v_b^m, i_a^{mc}, i_b^{mc}, v_a^c, v_b^c, i_a^{ck}, i_b^{ck}, v_a^k, v_b^k]^\top,$$

the four-dimensional input vector

$$\mathbf{u} = [i_a^m, i_b^m, i_a^k, i_b^k]^\top,$$

and the six-dimensional parameter vector

$$\mathbf{p} = [L, R, C, G, \omega, N]^\top.$$

Here  $N \in (0, 1)$  partitions the total shunt capacitance  $C$  and conductance  $G$  across the three  $\pi$ -sections via

$$\begin{aligned} C_1 &= \frac{1-N}{2} C, & C_2 &= N C, & C_3 &= \frac{1-N}{2} C, \\ G_1 &= \frac{1-N}{2} G, & G_2 &= N G, & G_3 &= \frac{1-N}{2} G, \end{aligned}$$

and each series branch has

$$R' = \frac{R}{2}, \quad L' = \frac{L}{2}.$$

*Section m (sending end):*

$$\dot{v}_a^m = -\frac{G_1}{C_1} v_a^m + \omega v_b^m + \frac{1}{C_1} (i_a^m - i_a^{mc}), \quad (10)$$

$$\dot{v}_b^m = -\frac{G_1}{C_1} v_b^m - \omega v_a^m + \frac{1}{C_1} (i_b^m - i_b^{mc}), \quad (11)$$

$$\dot{i}_a^{mc} = -\frac{R'}{L'} i_a^{mc} + \omega i_b^{mc} + \frac{1}{L'} (v_a^m - v_a^c), \quad (12)$$

$$\dot{i}_b^{mc} = -\frac{R'}{L'} i_b^{mc} - \omega i_a^{mc} + \frac{1}{L'} (v_b^m - v_b^c). \quad (13)$$

*Section c (central section):*

$$\dot{v}_a^c = -\frac{G_2}{C_2} v_a^c + \omega v_b^c + \frac{1}{C_2} (i_a^{mc} - i_a^{ck}), \quad (14)$$

$$\dot{v}_b^c = -\frac{G_2}{C_2} v_b^c - \omega v_a^c + \frac{1}{C_2} (i_b^{mc} - i_b^{ck}), \quad (15)$$

$$\dot{i}_a^{ck} = -\frac{R'}{L'} i_a^{ck} + \omega i_b^{ck} + \frac{1}{L'} (v_a^c - v_a^k), \quad (16)$$

$$\dot{i}_b^{ck} = -\frac{R'}{L'} i_b^{ck} - \omega i_a^{ck} + \frac{1}{L'} (v_b^c - v_b^k). \quad (17)$$

*Section k (receiving end):*

$$\dot{v}_a^k = -\frac{G_3}{C_3} v_a^k + \omega v_b^k + \frac{1}{C_3} (i_a^k + i_a^{ck}), \quad (18)$$

$$\dot{v}_b^k = -\frac{G_3}{C_3} v_b^k - \omega v_a^k + \frac{1}{C_3} (i_b^k + i_b^{ck}). \quad (19)$$

$$\mathbf{y} = [-i_a^{mc} \quad -i_b^{mc} \quad -i_a^{ck} \quad -i_b^{ck}]^T.$$

### C. Grid-Following Converter EMT Model

*1) Model Description:* The dynamics of the three-phase grid-following converter with LCL filter and PLL are captured by the fifteen-dimensional state vector

$$\mathbf{x} = [i_{c,a}, i_{c,b}, i_{c,c}, i_{g,a}, i_{g,b}, i_{g,c}, u_{f,a}, u_{f,b}, u_{f,c}, \quad (20)$$

$$\xi_{pll}, \theta_{pll}, \xi_P, \xi_Q, \xi_{id}, \xi_{iq}]^T, \quad (21)$$

the four-dimensional input vector

$$\mathbf{u} = [P^*, Q^*, E_{abs}, \phi_e]^T,$$

and the thirteen-dimensional parameter vector

$$\mathbf{p} = [L_c, L_g, C_f, R_c, R_f, R_{lg}, K_p^{pc}, K_i^{pc}, K_p^{cc}, K_i^{cc}, K_p^{pll}, K_i^{pll}, \omega_N]^T. \quad (22)$$

For compactness we define three-phase vectors

$$\mathbf{i}_c = \begin{bmatrix} i_{c,a} \\ i_{c,b} \\ i_{c,c} \end{bmatrix}, \quad \mathbf{i}_g = \begin{bmatrix} i_{g,a} \\ i_{g,b} \\ i_{g,c} \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}, \quad \mathbf{u}_f = \begin{bmatrix} u_{f,a} \\ u_{f,b} \\ u_{f,c} \end{bmatrix},$$

and similarly for the internal node voltages

$$\mathbf{u}_c = [u_{c,a}, u_{c,b}, u_{c,c}]^T, \quad \mathbf{u}_g = [u_{g,a}, u_{g,b}, u_{g,c}]^T, \quad (23)$$

$$\mathbf{u}_{conv} = [u_a, u_b, u_c]^T. \quad (24)$$

*2) Stiff Grid Source:* The three-phase stiff grid voltage is imposed as

$$\mathbf{e}(t) = E_{abs} \begin{bmatrix} \cos(\omega_N t + \phi_e) \\ \cos(\omega_N t + \phi_e - \frac{2\pi}{3}) \\ \cos(\omega_N t + \phi_e + \frac{2\pi}{3}) \end{bmatrix}. \quad (25)$$

A temporary amplitude reduction can be represented by

$$E_{abs}(t) = \begin{cases} 0.8 E_{abs}, & t \in (t_{fault}, t_{clear}), \\ E_{abs}, & \text{otherwise.} \end{cases}$$

*3) EMT Network Dynamics (LCL Filter):* The converter is connected to the grid through an LCL filter with converter-side inductance  $L_c$ , grid-side inductance  $L_g$ , damping resistors  $R_c$  and  $R_f$ , grid resistance  $R_{lg}$ , and a shunt capacitor  $C_f$  at the PCC. For each phase, the differential equations of the inductors and capacitor can be written in vector form as

$$L_c \dot{\mathbf{i}}_c = \mathbf{u}_{conv} - \mathbf{u}_c, \quad (26)$$

$$L_g \dot{\mathbf{i}}_g = \mathbf{v} - \mathbf{u}_g, \quad (27)$$

$$C_f \dot{\mathbf{u}}_f = \frac{\mathbf{v} - \mathbf{u}_f}{R_f}. \quad (28)$$

The resistive and algebraic constraints follow from Ohm's law and KCL at the internal nodes:

$$\mathbf{0} = \mathbf{i}_c + \frac{\mathbf{u}_c - \mathbf{v}}{R_c}, \quad (29)$$

$$\mathbf{0} = \mathbf{i}_g + \frac{\mathbf{u}_g - \mathbf{e}(t)}{R_{lg}}, \quad (30)$$

$$\mathbf{0} = \frac{\mathbf{v} - \mathbf{u}_c}{R_c} + \frac{\mathbf{v} - \mathbf{u}_f}{R_f} + \mathbf{i}_g. \quad (31)$$

*4) PLL Dynamics:* The phase-locked loop is modeled in the synchronous  $dq$  frame aligned with the PCC voltage. Its states are the integrator  $\xi_{pll}$  and the estimated grid angle  $\theta_{pll}$ . With  $v_q$  the  $q$ -axis PCC voltage, the PLL dynamics are

$$\dot{\xi}_{pll} = v_q, \quad (32)$$

$$\dot{\theta}_{pll} = K_p^{pll} v_q + K_i^{pll} \xi_{pll} + \omega_N. \quad (33)$$

*5) Outer Power Control:* The injected active and reactive powers at the PCC are computed from the  $dq$ -axis voltages and currents as

$$P = v_d i_d + v_q i_q, \quad (34)$$

$$Q = v_q i_d - v_d i_q. \quad (35)$$

The outer control loop regulates these quantities to the references  $P^*$  and  $Q^*$ . Power errors are

$$dP = P^* - P, \quad (36)$$

$$dQ = Q - Q^*, \quad (37)$$

and a pair of PI controllers with integrator states  $\xi_P$  and  $\xi_Q$  generate the current references

$$\dot{\xi}_P = dP, \quad (38)$$

$$i_d^* = K_p^{pc} dP + K_i^{pc} \xi_P, \quad (39)$$

$$\dot{\xi}_Q = dQ, \quad (40)$$

$$i_q^* = K_p^{pc} dQ + K_i^{pc} \xi_Q. \quad (41)$$

6) *Inner Current Control*: In the inner loop, the  $d$ - and  $q$ -axis current errors are

$$e_{id} = i_d^* - i_d, \quad e_{iq} = i_q^* - i_q,$$

and are processed by PI controllers with integrator states  $\xi_{id}$  and  $\xi_{iq}$ :

$$\dot{\xi}_{id} = e_{id}, \quad (42)$$

$$u_d = K_p^{cc} e_{id} + K_i^{cc} \xi_{id}, \quad (43)$$

$$\dot{\xi}_{iq} = e_{iq}, \quad (44)$$

$$u_q = K_p^{cc} e_{iq} + K_i^{cc} \xi_{iq}. \quad (45)$$

7) *Park Transformations and Converter Voltage Synthesis*: Using  $\theta_{pll}$  as the synchronous angle, the abc-to- $dq$  Park transformation for the PCC voltages is

$$\theta_a = \theta_{pll}, \quad \theta_b = \theta_{pll} - \frac{2\pi}{3}, \quad \theta_c = \theta_{pll} + \frac{2\pi}{3}. \quad (46)$$

The PCC voltages are transformed to the synchronous  $dq$  frame as

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \frac{2}{3} \sum_{p \in \{a,b,c\}} v_p \begin{bmatrix} \cos \theta_p \\ -\sin \theta_p \end{bmatrix}. \quad (47)$$

Similarly, the grid currents are transformed according to

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \frac{2}{3} \sum_{p \in \{a,b,c\}} i_{g,p} \begin{bmatrix} \cos \theta_p \\ -\sin \theta_p \end{bmatrix}. \quad (48)$$

The inverse Park transformation maps the  $dq$ -axis voltage commands to converter phase voltages:

$$u_a = \cos \theta_{pll} u_d - \sin \theta_{pll} u_q, \quad (49)$$

$$u_b = \cos(\theta_{pll} - \frac{2\pi}{3}) u_d - \sin(\theta_{pll} - \frac{2\pi}{3}) u_q, \quad (50)$$

$$u_c = \cos(\theta_{pll} + \frac{2\pi}{3}) u_d - \sin(\theta_{pll} + \frac{2\pi}{3}) u_q. \quad (51)$$

8) *DAE Formulation*: Collecting all terms, the model can be written compactly as a nonlinear differential-algebraic system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \mathbf{p}, t), \quad (52)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{u}, \mathbf{p}, t), \quad (53)$$

where  $\mathbf{f}$  is defined by the differential equations (26)–(28), (32)–(33), and the PI controller dynamics for  $\xi_P, \xi_Q, \xi_{id}, \xi_{iq}$ , while  $\mathbf{g}$  collects the algebraic constraints (29)–(31), the power expressions, and the abc- $dq$  mappings.

#### D. Synchronous Machine-Infinite Bus EMT Model

1) *Model Description*: The dynamics of the synchronous machine connected to an infinite bus through a three-phase transmission element are captured by the nineteen-dimensional state vector

$$\mathbf{x} = [i_a, i_b, i_c, \psi_d, \psi_q, \psi_f, \psi_{q1}, \psi_o, v_a, v_b, v_c, \quad (54)$$

$$v_d, v_q, v_o, i_d, i_q, i_o, i_f, i_{q1}]^T, \quad (55)$$

where  $\mathbf{i} = [i_a, i_b, i_c]^T$  are stator phase currents,  $\psi_d, \psi_q$  are rotor  $dq$ -axis stator flux linkages,  $\psi_f$  and  $\psi_{q1}$  are field and damper winding flux linkages,  $\psi_o$  is the zero-sequence flux,  $\mathbf{v} = [v_a, v_b, v_c]^T$  are the terminal phase voltages, and

$(v_d, v_q, v_o)$  and  $(i_d, i_q, i_o)$  are the synchronous-frame voltage and current components.

The infinite bus acts as a stiff Thevenin source

$$\mathbf{e}(t) = \sqrt{2} |E_\infty| \begin{bmatrix} \cos(\omega_N t + \phi_E) \\ \cos(\omega_N t + \phi_E - 2\pi/3) \\ \cos(\omega_N t + \phi_E + 2\pi/3) \end{bmatrix}, \quad (56)$$

with a temporary outage represented by setting  $\mathbf{e}(t) = \mathbf{0}$  over prescribed time intervals.

*Network (Line) Current Dynamics*: Each stator phase current flows through a per-phase series impedance  $(R_e, L_e)$ . For  $p \in \{a, b, c\}$ ,

$$\dot{i}_p = \omega_N \left( -\frac{R_e}{L_e} i_p + \frac{1}{L_e} v_p - \frac{1}{L_e} e_p(t) \right). \quad (57)$$

*Synchronous Machine Flux Dynamics*: The machine's  $dq0$  flux derivatives follow directly from the Park-transformed stator equations and the field and damper winding circuits:

$$\dot{\psi}_d = \omega_N (-R_a i_d - \psi_q - v_d), \quad (58)$$

$$\dot{\psi}_q = \omega_N (-R_a i_q + \psi_d - v_q), \quad (59)$$

$$\dot{\psi}_f = \omega_N (-R_f i_f + V_f), \quad (60)$$

$$\dot{\psi}_{q1} = \omega_N (-R_{q1} i_{q1}), \quad (61)$$

$$\dot{\psi}_o = \omega_N (-R_a i_o - v_o). \quad (62)$$

*abc- $dq0$  Transformations*: Let  $\theta(t) = \theta_0 + \omega_N t$  denote the electrical rotor angle. The terminal voltages must satisfy the Park transformation constraints:

$$0 = \frac{\sqrt{2}}{3} (\cos \theta v_a + \cos(\theta - 2\pi/3) v_b + \cos(\theta + 2\pi/3) v_c) - v_d, \quad (63)$$

$$0 = -\frac{\sqrt{2}}{3} (\sin \theta v_a + \sin(\theta - 2\pi/3) v_b + \sin(\theta + 2\pi/3) v_c) - v_q, \quad (64)$$

$$0 = \frac{1}{3} (v_a + v_b + v_c) - v_o. \quad (65)$$

Similarly, the stator currents must satisfy

$$0 = \frac{\sqrt{2}}{3} (\cos \theta i_a + \cos(\theta - 2\pi/3) i_b + \cos(\theta + 2\pi/3) i_c) - i_d, \quad (66)$$

$$0 = -\frac{\sqrt{2}}{3} (\sin \theta i_a + \sin(\theta - 2\pi/3) i_b + \sin(\theta + 2\pi/3) i_c) - i_q, \quad (67)$$

$$0 = \frac{1}{3} (i_a + i_b + i_c) - i_o. \quad (68)$$

*Machine Algebraic Equations*: Flux linkages and currents obey the magnetic coupling relations

$$0 = L_{dd} i_d + L_{df} i_f - \psi_d, \quad (69)$$

$$0 = L_{qq} i_q + L_{q1} i_{q1} - \psi_q, \quad (70)$$

$$0 = L_{ff} i_f + L_{df} i_d - \psi_f, \quad (71)$$

$$0 = L_{q1} i_{q1} + L_{qq1} i_q - \psi_{q1}, \quad (72)$$

$$0 = L_{oo} i_o - \psi_o. \quad (73)$$

*DAE Formulation:* Collecting the differential and algebraic equations above, the synchronous machine–infinite bus model satisfies the nonlinear DAE system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \quad (74)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}, t), \quad (75)$$

where the functions  $\mathbf{f}$  and  $\mathbf{g}$  contain, respectively, the line-current dynamics, synchronous-machine flux dynamics, and the abc–dq0 algebraic constraints together with the magnetic coupling equations.

#### E. Networked IEEE RMS Benchmark Systems

The networked RMS benchmarks are constructed from the IEEE 9-, 14-, 39-, 57-, and 118-bus transmission systems. The original AC network of each test system is retained, including all buses, branches, transformers, shunt elements, generation buses, and loads. The steady-state operating point is determined from an AC power flow and provides the initial condition for the dynamic simulation.

The IEEE network data specify steady-state electrical quantities rather than a unique set of dynamic parameters. We therefore augment the steady-state networks with a common dynamic benchmark parameterization. The same component models and parameterization rules are used for all five systems. The resulting benchmarks should consequently be interpreted as dynamic RMS systems constructed on the standard IEEE networks, rather than as historical dynamic-data sets associated with the original IEEE cases.

Throughout this appendix, quantities are expressed on a  $S_{\text{base}} = 100$  MVA system base and the nominal electrical frequency is

$$f_b = 60 \text{ Hz}, \quad \omega_b = 2\pi f_b. \quad (76)$$

Each system contains three classes of dynamic devices: sixth-order synchronous machines with excitation and governor–turbine dynamics, grid-following inverter-based resources with PLL and power-control dynamics, and fifth-order induction-motor aggregates. The remainder of the demand is represented by static impedance. The complete differential state is therefore

$$\mathbf{x} = [\mathbf{x}_{\text{SG}}^T \quad \mathbf{x}_{\text{IBR}}^T \quad \mathbf{x}_{\text{IM}}^T]^T. \quad (77)$$

For  $N_{\text{SG}}$  synchronous machines,  $N_{\text{IBR}}$  grid-following inverters, and  $N_{\text{IM}}$  induction motors, the number of differential states is

$$n_x = 10N_{\text{SG}} + 8N_{\text{IBR}} + 5N_{\text{IM}}. \quad (78)$$

The resulting composition of the five benchmark systems is summarized in Table I.

A target fraction

$$\eta_{\text{IBR}} = 0.20 \quad (79)$$

of the dispatched system active power is represented by grid-following IBRs. Since the steady-state IEEE cases do not specify generation technology, the allocation is performed at existing generator buses. Active generating units are ordered by dispatched active power and the smallest units are assigned to IBR operation first until the target system-wide IBR share

is reached. A marginal unit is split when required to obtain exactly  $\eta_{\text{IBR}} = 20\%$ . Zero-active-power synchronous condensers remain synchronous devices.

For the IEEE 9-bus system, bus 1 is the marginal plant and its IBR fraction is 0.8893. For IEEE 14, bus 2 is entirely inverter based and bus 1 has an IBR fraction of 0.0623. For IEEE 39, buses 30 and 34 are entirely inverter based, while the IBR fraction at bus 37 is 0.9288. For IEEE 57, bus 3 is entirely inverter based and the IBR fraction at bus 12 is 0.6959. For IEEE 118, all listed IBR plants except bus 25 are entirely represented by the grid-following model; bus 25 is the marginal plant with IBR fraction 0.5313.

At selected load buses, the target active-power fraction assigned to the induction-motor aggregate is

$$\eta_{\text{IM}} = 0.25. \quad (80)$$

At most six dynamic motor buses are used. Candidate buses satisfy

$$P_{L,k} > 0.03 \text{ pu}, \quad Q_{L,k} > 0.005 \text{ pu}, \quad (81)$$

and are ordered by active demand. The motor fraction is additionally limited when necessary to avoid assigning more reactive demand to the motor than is available at the original operating point. Consequently, the resulting system-wide dynamic-motor active-power fractions are those reported in Table I. The remaining load is retained as static impedance.

The complete networked model is a semi-explicit index-1 DAE,

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}, \mathbf{V}, \sigma), \quad (82a)$$

$$\mathbf{0} = \mathbf{A}_\sigma \mathbf{V} - \mathbf{b}(\mathbf{x}), \quad (82b)$$

where  $\mathbf{V} \in \mathbb{C}^{n_b}$  contains the complex positive-sequence bus voltages and  $\sigma$  denotes the active network or disturbance discrete state.

The second equation in (82) is an algebraic Kirchhoff-current constraint on the complete network. It does not represent a reduction in the number of buses. Since  $\mathbf{A}_\sigma$  is nonsingular in the considered operating region, the algebraic voltage solution is

$$\mathbf{V} = \mathbf{A}_\sigma^{-1} \mathbf{b}(\mathbf{x}). \quad (83)$$

Thus, the bus voltages can be eliminated algebraically while retaining the complete physical network and all differential component states.

The initial operating point is obtained from the conventional AC power-flow equations. Let

$$V_k = |V_k| e^{j\theta_k}, \quad (84)$$

and let  $\mathbf{Y}_{\text{bus}}$  denote the nodal admittance matrix of the corresponding IEEE network. The complex power injected at bus  $k$  is

$$S_k = P_k + jQ_k = V_k \left( \sum_{\ell=1}^{n_b} Y_{k\ell} V_\ell \right)^*. \quad (85)$$

The slack-bus angle, PV-bus voltage magnitudes, specified active-power injections, and PQ-bus active and reactive powers

TABLE I: Composition of the IEEE networked RMS benchmarks.

| System   | $n_b$ | Branches | Gen. records | $N_{SG}$ | $N_{IBR}$ | $N_{IM}$ | $n_x$ | $P_{IM}/P_L$ | IBR buses                                    | Motor buses            |
|----------|-------|----------|--------------|----------|-----------|----------|-------|--------------|----------------------------------------------|------------------------|
| IEEE 9   | 9     | 9        | 3            | 3        | 1         | 3        | 53    | 25.00%       | 1                                            | 5, 7, 9                |
| IEEE 14  | 14    | 20       | 5            | 4        | 2         | 6        | 86    | 17.86%       | 1, 2                                         | 2, 3, 6, 9, 13, 14     |
| IEEE 39  | 39    | 46       | 10           | 8        | 3         | 6        | 134   | 11.72%       | 30, 34, 37                                   | 3, 4, 8, 16, 20, 39    |
| IEEE 57  | 57    | 80       | 7            | 6        | 2         | 6        | 106   | 9.49%        | 3, 12                                        | 1, 6, 8, 9, 12, 16     |
| IEEE 118 | 118   | 186      | 54           | 44       | 11        | 6        | 558   | 5.12%        | 12, 25, 31, 46, 49, 54, 59, 61, 87, 103, 111 | 15, 42, 54, 59, 80, 90 |

define the usual power-flow constraints. The converged solution  $\mathbf{V}_0$  is subsequently used to initialize each dynamic device such that its initial electrical injection is consistent with the power-flow solution.

Each synchronous generator is represented by

$$\mathbf{x}_g = \begin{bmatrix} \delta_g \\ \omega_g \\ E'_{q,g} \\ E'_{d,g} \\ E''_{q,g} \\ E''_{d,g} \\ E_{fd,g} \\ V_{m,g} \\ P_{m,g} \\ P_{v,g} \end{bmatrix} \in \mathbb{R}^{10}. \quad (86)$$

Here  $\delta_g$  is the rotor electrical angle,  $\omega_g$  the rotor speed in per unit,  $E'_{q,g}$  and  $E'_{d,g}$  the transient internal voltages,  $E''_{q,g}$  and  $E''_{d,g}$  the subtransient internal voltages,  $E_{fd,g}$  the excitation voltage,  $V_{m,g}$  the measured terminal-voltage magnitude,  $P_{m,g}$  the mechanical power, and  $P_{v,g}$  the governor/valve state.

For a generator connected at bus  $k$ , define

$$\gamma_g = \delta_g - \frac{\pi}{2}, \quad (87)$$

and express its terminal voltage in the rotor reference frame as

$$V_{d,g} + jV_{q,g} = V_k e^{-j\gamma_g}. \quad (88)$$

The subtransient stator currents are

$$I_{d,g} = \frac{E''_{q,g} - V_{q,g}}{x''_g}, \quad (89)$$

$$I_{q,g} = \frac{V_{d,g} - E''_{d,g}}{x''_g}. \quad (90)$$

The current injected into the network is

$$I_g = (I_{d,g} + jI_{q,g})e^{j\gamma_g}, \quad (91)$$

and the corresponding electrical active power is

$$P_{e,g} = \Re\{V_k I_g^*\}. \quad (92)$$

The electromechanical rotor equations are

$$\dot{\delta}_g = \omega_b(\omega_g - 1), \quad (93)$$

$$\dot{\omega}_g = \frac{P_{m,g} - P_{e,g} - D_g(\omega_g - 1)}{2H_g}. \quad (94)$$

The transient and subtransient voltage dynamics are

$$T'_{do,g} \dot{E}'_{q,g} = E_{fd,g} - E'_{q,g} - (x_{d,g} - x'_{d,g})I_{d,g}, \quad (95)$$

$$T'_{qo,g} \dot{E}'_{d,g} = -E'_{d,g} + (x_{q,g} - x'_{q,g})I_{q,g}, \quad (96)$$

$$T''_{do,g} \dot{E}''_{q,g} = E'_{q,g} - E''_{q,g} - (x'_{d,g} - x''_{d,g})I_{d,g}, \quad (97)$$

$$T''_{qo,g} \dot{E}''_{d,g} = E'_{d,g} - E''_{d,g} + (x'_{q,g} - x''_{q,g})I_{q,g}. \quad (98)$$

The terminal-voltage measurement is represented by

$$T_{r,g} \dot{V}_{m,g} = |V_k| - V_{m,g}. \quad (99)$$

The excitation command is

$$E_{fd,g}^* = \text{clip}(K_{A,g}[V_{\text{ref},g} - V_{m,g}], -100, 100), \quad (100)$$

followed by

$$T_{A,g} \dot{E}_{fd,g} = E_{fd,g}^* - E_{fd,g}. \quad (101)$$

The wide bound in (100) acts as a soft field runaway guard and remains inactive close to the initialized operating point.

The governor and turbine are represented by

$$T_{g,g} \dot{P}_{v,g} = P_{\text{ref},g} - \frac{\omega_g - 1}{R_g} - P_{v,g}, \quad (102)$$

$$T_{t,g} \dot{P}_{m,g} = P_{v,g} - P_{m,g}. \quad (103)$$

The synchronous-machine parameters common to the benchmark are

$$\begin{aligned} x_d &= 1.80, & x_q &= 1.70, & x'_d &= 0.30, & x'_q &= 0.55, \\ T'_{qo} &= 0.40 \text{ s}, & T''_{do} &= 0.030 \text{ s}, & T''_{qo} &= 0.050 \text{ s}, & D &= 1.0, \\ K_A &= 20, & T_A &= 0.050 \text{ s}, & T_r &= 0.020 \text{ s}, & R_g &= 0.05, \\ T_g &= 0.20 \text{ s}, & T_t &= 0.50 \text{ s}. \end{aligned} \quad (104)$$

To avoid an artificially identical machine population, three machine parameters vary deterministically with the dispatched-unit index  $r = 0, 1, \dots$ :

$$H_r = 3.5 + 0.4[(r \bmod 4) + 1] \text{ s}, \quad (105)$$

$$T'_{do,r} = 6.0 + 0.7(r \bmod 4) \text{ s}, \quad (106)$$

$$x''_r = 0.20 + 0.01(r \bmod 3). \quad (107)$$

Consequently,

$$H_r \in \{3.9, 4.3, 4.7, 5.1\} \text{ s}, \quad (108)$$

$$T'_{do,r} \in \{6.0, 6.7, 7.4, 8.1\} \text{ s}, \quad (109)$$

and

$$x''_r \in \{0.20, 0.21, 0.22\}. \quad (110)$$

At initialization, the machine states are selected to satisfy the power-flow current injection. If  $S_g$  and  $V_k$  denote the

assigned initial complex power and terminal voltage, respectively,

$$I_{g,0} = \left( \frac{S_g}{V_k} \right)^* . \quad (111)$$

The initial internal subtransient voltage is

$$E''_{g,0} = V_k + jx''_g I_{g,0}, \quad (112)$$

from which  $\delta_g$ ,  $E''_{q,g}$ ,  $E''_{d,g}$ ,  $E'_{q,g}$ ,  $E'_{d,g}$ , and  $E_{fd,g}$  are obtained. The mechanical states are initialized according to

$$P_{m,g}(0) = P_{v,g}(0) = \Re\{S_g\}, \quad (113)$$

and

$$V_{\text{ref},g} = |V_k| + \frac{E_{fd,g}(0)}{K_A}, \quad (114)$$

so that the machine starts from the power-flow equilibrium.

Each grid-following IBR is represented by the state vector

$$\mathbf{x}_i = [\theta_i \quad \xi_{\text{pll},i} \quad P_{f,i} \quad Q_{f,i} \quad \xi_{P,i} \quad \xi_{Q,i} \quad i_{d,i} \quad i_{q,i}]^T. \quad (115)$$

The states describe the estimated grid angle, PLL integral state, filtered active and reactive powers, outer-loop PI integral states, and the  $dq$  current injection.

For an inverter connected to bus  $k$ , its rotating-frame terminal voltage is

$$v_{d,i} + jv_{q,i} = V_k e^{-j\theta_i}. \quad (116)$$

The injected active and reactive powers are

$$P_i = v_{d,i}i_{d,i} + v_{q,i}i_{q,i}, \quad (117)$$

$$Q_i = v_{q,i}i_{d,i} - v_{d,i}i_{q,i}. \quad (118)$$

The normalized PLL error is

$$e_{\text{pll},i} = \frac{v_{q,i}}{\max(|V_k|, V_\epsilon)}, \quad (119)$$

with

$$\dot{\xi}_{\text{pll},i} = e_{\text{pll},i}, \quad (120)$$

$$\dot{\theta}_i = k_{p,\text{pll}} e_{\text{pll},i} + k_{i,\text{pll}} \xi_{\text{pll},i}. \quad (121)$$

The measured active and reactive powers are filtered according to

$$T_{pq} \dot{P}_{f,i} = P_i - P_{f,i}, \quad (122)$$

$$T_{pq} \dot{Q}_{f,i} = Q_i - Q_{f,i}. \quad (123)$$

Defining

$$e_{P,i} = P_{\text{ref},i} - P_{f,i}, \quad e_{Q,i} = Q_{\text{ref},i} - Q_{f,i}, \quad (124)$$

the outer PI integrators satisfy

$$\dot{\xi}_{P,i} = e_{P,i}, \quad (125)$$

$$\dot{\xi}_{Q,i} = e_{Q,i}. \quad (126)$$

The voltage used to form the current request is regularized as

$$v_{d,\epsilon} = \max(|v_{d,i}|, V_\epsilon), \quad (127)$$

where

$$V_\epsilon = 0.20 \text{ pu}. \quad (128)$$

The unconstrained current references are then

$$i_{d,i}^{\star,\text{raw}} = \frac{P_{\text{ref},i}}{v_{d,\epsilon}} + k_{p,P} e_{P,i} + k_{i,P} \xi_{P,i}, \quad (129)$$

$$i_{q,i}^{\star,\text{raw}} = -\frac{Q_{\text{ref},i}}{v_{d,\epsilon}} - k_{p,Q} e_{Q,i} - k_{i,Q} \xi_{Q,i}. \quad (130)$$

A circular converter-current limit is applied to these commands. Define

$$I_i^{\star,\text{raw}} = \sqrt{(i_{d,i}^{\star,\text{raw}})^2 + (i_{q,i}^{\star,\text{raw}})^2}, \quad (131)$$

and

$$\kappa_i = \min \left( 1, \frac{I_{\text{max},i}}{\max(I_i^{\star,\text{raw}}, \epsilon)} \right). \quad (132)$$

The limited commands are

$$\begin{bmatrix} i_{d,i}^{\star} \\ i_{q,i}^{\star} \end{bmatrix} = \kappa_i \begin{bmatrix} i_{d,i}^{\star,\text{raw}} \\ i_{q,i}^{\star,\text{raw}} \end{bmatrix}, \quad (133)$$

which guarantees

$$\sqrt{(i_{d,i}^{\star})^2 + (i_{q,i}^{\star})^2} \leq I_{\text{max},i}. \quad (134)$$

The inner converter/current response is represented by a first-order lag,

$$T_i \dot{i}_{d,i} = i_{d,i}^{\star} - i_{d,i}, \quad (135)$$

$$T_i \dot{i}_{q,i} = i_{q,i}^{\star} - i_{q,i}. \quad (136)$$

The current injected into the network is

$$I_{\text{IBR},i} = (i_{d,i} + ji_{q,i})e^{j\theta_i}. \quad (137)$$

The common IBR parameters are

$$\begin{aligned} k_{p,\text{pll}} &= 35, & k_{i,\text{pll}} &= 600, & T_{pq} &= 0.020 \text{ s}, \\ k_{p,P} &= 0.40, & k_{i,P} &= 15, & k_{p,Q} &= 0.40, \\ k_{i,Q} &= 15, & T_i &= 0.004 \text{ s}, & V_\epsilon &= 0.20 \text{ pu}. \end{aligned} \quad (138)$$

The initial inverter angle is aligned with the power-flow voltage,

$$\theta_i(0) = \angle V_k, \quad (139)$$

and the initial  $dq$  current is obtained from

$$I_{i,0} = \left( \frac{S_i}{V_k} \right)^*, \quad I_{dq,i,0} = I_{i,0} e^{-j\theta_i(0)}. \quad (140)$$

The references are initialized as

$$P_{\text{ref},i} = P_i(0), \quad Q_{\text{ref},i} = Q_i(0), \quad (141)$$

with

$$P_{f,i}(0) = P_i(0), \quad Q_{f,i}(0) = Q_i(0). \quad (142)$$

The current capability is scaled to the initialized device injection:

$$I_{\text{max},i} = 1.30 \max(|I_{dq,i,0}|, 0.10). \quad (143)$$

Thus, an initialized inverter with appreciable power injection has approximately 30% current headroom relative to its initial operating current.

Each dynamic induction-motor aggregate is represented by

$$\mathbf{x}_m = [E'_{q,m} \quad E'_{d,m} \quad E''_{q,m} \quad E''_{d,m} \quad s_m]^T, \quad (144)$$

where  $s_m$  is rotor slip. For the positive-sequence terminal voltage

$$V_k = V_d + jV_q, \quad (145)$$

define

$$\Delta_m = r_s^2 + (L'')^2. \quad (146)$$

The motor currents are

$$i_{d,m} = \frac{r_s(V_d + E''_{d,m}) + L''(V_q + E''_{q,m})}{\Delta_m}, \quad (147)$$

$$i_{q,m} = \frac{r_s(V_q + E''_{q,m}) - L''(V_d + E''_{d,m})}{\Delta_m}. \quad (148)$$

The transient electrical dynamics satisfy

$$T' \dot{E}'_{q,m} = -E'_{q,m} - (L_s - L')i_{d,m} - \omega_b s_m T' E'_{d,m}, \quad (149)$$

$$T' \dot{E}'_{d,m} = -E'_{d,m} + (L_s - L')i_{q,m} + \omega_b s_m T' E'_{q,m}. \quad (150)$$

Define

$$a_m = \frac{T''(L_s - L') + T'(L' - L'')}{T'T''}. \quad (151)$$

The subtransient dynamics are

$$\dot{E}''_{q,m} = \frac{T' - T''}{T'T''} E'_{q,m} - a_m i_{d,m} - \frac{E''_{q,m}}{T''} - \omega_b s_m E''_{d,m}, \quad (152)$$

$$\dot{E}''_{d,m} = \frac{T' - T''}{T'T''} E'_{d,m} + a_m i_{q,m} - \frac{E''_{d,m}}{T''} + \omega_b s_m E''_{q,m}. \quad (153)$$

The mechanical rotor speed is

$$\omega_m = 1 - s_m. \quad (154)$$

The electrical air-gap power is

$$P_{\text{gap},m} = \Re \{V_k I_m^*\} - r_s |I_m|^2, \quad (155)$$

where

$$I_m = i_{d,m} + j i_{q,m}. \quad (156)$$

The electromagnetic torque is

$$T_{e,m} = \frac{P_{\text{gap},m}}{\max(\omega_m, 0.05)}. \quad (157)$$

The mechanical load torque follows the speed-dependent characteristic

$$T_{m,m} = T_{m0,m} \mu_m(t) [\max(\omega_m, 0.05)]^{E_{\text{trq}}}, \quad (158)$$

and the slip equation is

$$\dot{s}_m = \frac{T_{m,m} - T_{e,m}}{2H_m}. \quad (159)$$

The common induction-motor parameters are

$$\begin{aligned} r_s &= 0.020, & L_s &= 2.50, & L' &= 0.18, & L'' &= 0.12, \\ T' &= 0.15 \text{ s}, & T'' &= 0.025 \text{ s}, & H_m &= 0.50 \text{ s}, & E_{\text{trq}} &= 2.0. \end{aligned} \quad (160)$$

The initial slip is

$$s_m(0) = 0.04, \quad \omega_m(0) = 0.96. \quad (161)$$

For the prescribed initial slip and power-flow voltage, the four electrical states are obtained from their steady-state equations,

$$\dot{E}'_q = \dot{E}'_d = \dot{E}''_q = \dot{E}''_d = 0. \quad (162)$$

The initial electromagnetic torque is then used to determine  $T_{m0,m}$  such that

$$\dot{s}_m(0) = 0. \quad (163)$$

For each selected motor bus, the motor model is first evaluated with unit scaling. If its corresponding active power is  $P_{m,\text{unit}}$ , the motor scale is selected to target

$$P_{m,\text{target}} = 0.25 P_{L,k}, \quad (164)$$

subject to the available reactive-power demand at that bus. The remaining complex load is represented by static impedance.

For a static load with initial complex power

$$S_{Z,k} = P_{Z,k} + jQ_{Z,k} \quad (165)$$

and initial bus voltage  $V_{k,0}$ , the equivalent shunt admittance is

$$Y_{Z,k} = \frac{S_{Z,k}^*}{|V_{k,0}|^2}. \quad (166)$$

Its current is therefore

$$I_{Z,k} = Y_{Z,k} V_k, \quad (167)$$

so its active and reactive consumption vary with voltage magnitude rather than remaining constant after a disturbance.

All components are coupled through the complete positive-sequence AC network. Let

$$\mathbf{Y}_{\text{bus}} \in \mathbb{C}^{n_b \times n_b} \quad (168)$$

denote the bus-admittance matrix assembled from the IEEE branches, transformers, charging susceptances, and bus shunts.

Each synchronous generator admits a Norton representation. Define its subtransient internal voltage in network coordinates as

$$E''_g = (E''_{d,g} + jE''_{q,g})e^{j(\delta_g - \pi/2)}. \quad (169)$$

Its shunt contribution is

$$Y_{\text{SG},g} = \frac{1}{jx''_g}, \quad (170)$$

and its Norton source is

$$b_g = \frac{E''_g}{jx''_g}. \quad (171)$$

The GFL inverter behaves as a controlled current source,

$$b_i = (i_{d,i} + j i_{q,i})e^{j\theta_i}. \quad (172)$$

For induction motor  $m$ , define

$$E''_m = E''_{d,m} + jE''_{q,m}. \quad (173)$$

Its network admittance is

$$Y_{\text{IM},m} = \frac{\kappa_m}{r_s + jL''}, \quad (174)$$

where  $\kappa_m$  denotes the motor power scaling at its bus, and its Norton source is

$$b_m = -\frac{\kappa_m E''_m}{r_s + jL''}. \quad (175)$$

Collecting all static and dynamic admittances yields

$$\mathbf{A} = \mathbf{Y}_{\text{bus}} + \text{diag}(\mathbf{Y}_Z + \mathbf{Y}_{\text{SG}} + \mathbf{Y}_{\text{IM}}), \quad (176)$$

while the source at bus  $k$  is

$$b_k(\mathbf{x}) = \sum_{g \in \mathcal{G}_k} \frac{(E''_{d,g} + jE''_{q,g})e^{j(\delta_g - \pi/2)}}{jx''_g} + \sum_{i \in \mathcal{I}_k} (i_{d,i} + ji_{q,i})e^{j\theta_i} - \sum_{m \in \mathcal{M}_k} \frac{\kappa_m(E''_{d,m} + jE''_{q,m})}{r_s + jL''}. \quad (177)$$

Here  $\mathcal{G}_k$ ,  $\mathcal{I}_k$ , and  $\mathcal{M}_k$  denote the sets of synchronous machines, inverters, and motors connected at bus  $k$ , respectively.

Kirchhoff's current law for all buses can therefore be written compactly as

$$\mathbf{AV} = \mathbf{b}(\mathbf{x}). \quad (178)$$

Equation (178) is linear in the algebraic bus voltages, although  $\mathbf{b}(\mathbf{x})$  depends nonlinearly on the dynamic states. Hence,

$$\mathbf{V} = \mathbf{A}^{-1}\mathbf{b}(\mathbf{x}), \quad (179)$$

and substitution into the component equations gives the equivalent reduced differential representation

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, \mathbf{A}^{-1}\mathbf{b}(\mathbf{x}), t). \quad (180)$$

This elimination is algebraically exact for the specified model and does not constitute a Kron reduction of the IEEE network.

Two disturbance families are used. In the mixed-setpoint scenario, three successive disturbances excite the motor, inverter, and synchronous-machine control dynamics. With disturbance-scale parameter  $d$  and nominal value

$$d = 1, \quad (181)$$

the induction-motor mechanical-torque multiplier is

$$\mu_m(t) = \begin{cases} 1, & t < 0.20 \text{ s}, \\ 1 + 0.05d, & t \geq 0.20 \text{ s}. \end{cases} \quad (182)$$

The inverter active-power reference multiplier is

$$\mu_{\text{IBR}}(t) = \begin{cases} 1, & t < 0.55 \text{ s}, \\ 1 + 0.03d, & t \geq 0.55 \text{ s}, \end{cases} \quad (183)$$

such that

$$P_{\text{ref},i}(t) = \mu_{\text{IBR}}(t)P_{\text{ref},i}^0. \quad (184)$$

Finally, the synchronous-machine voltage reference is

$$V_{\text{ref},g}(t) = \begin{cases} V_{\text{ref},g}^0, & t < 0.80 \text{ s}, \\ V_{\text{ref},g}^0 + 0.01d, & t \geq 0.80 \text{ s}. \end{cases} \quad (185)$$

A separate fault-ride-through scenario introduces a balanced positive-sequence three-phase fault at the bus with the largest active load. The fault is represented by the finite shunt impedance

$$Z_f = 0.01 + j0.10 \text{ pu}, \quad (186)$$

or equivalently

$$Y_f = \frac{1}{0.01 + j0.10}. \quad (187)$$

The fault is active over

$$0.20 \text{ s} \leq t < 0.28 \text{ s}. \quad (188)$$

During this interval, the network matrix becomes

$$\mathbf{A}_{\text{fault}} = \mathbf{A} + \text{diag}(\mathbf{y}_f), \quad (189)$$

where  $\mathbf{y}_f$  is zero at all buses except the faulted bus, where its value is  $Y_f$ . The fault is cleared at  $t = 0.28 \text{ s}$ , after which the pre-fault network admittance is restored.

In the fault-ride-through scenario, the IBR active-power references are also increased at

$$t = 0.65 \text{ s} \quad (190)$$

according to

$$P_{\text{ref},i}(t) = \begin{cases} P_{\text{ref},i}^0, & t < 0.65 \text{ s}, \\ (1 + 0.02d)P_{\text{ref},i}^0, & t \geq 0.65 \text{ s}. \end{cases} \quad (191)$$

The differential states remain continuous across network events, whereas the algebraic voltages may change instantaneously because the active algebraic constraint changes from  $\mathbf{AV} = \mathbf{b}(\mathbf{x})$  to  $\mathbf{A}_{\text{fault}}\mathbf{V} = \mathbf{b}(\mathbf{x})$  and back. Setpoint disturbances change the forcing of the dynamic equations without changing the network topology.

Collecting the complete component models, every benchmark therefore has the common structure

$$\dot{\mathbf{x}}_{\text{SG}} = \mathbf{f}_{\text{SG}}(\mathbf{x}_{\text{SG}}, \mathbf{V}, t), \quad (192a)$$

$$\dot{\mathbf{x}}_{\text{IBR}} = \mathbf{f}_{\text{IBR}}(\mathbf{x}_{\text{IBR}}, \mathbf{V}, t), \quad (192b)$$

$$\dot{\mathbf{x}}_{\text{IM}} = \mathbf{f}_{\text{IM}}(\mathbf{x}_{\text{IM}}, \mathbf{V}, t), \quad (192c)$$

$$\mathbf{0} = \mathbf{A}_\sigma \mathbf{V} - \mathbf{b}(\mathbf{x}). \quad (192d)$$

The five systems therefore differ primarily in network size, generation dispatch, dynamic-device placement, and the resulting state dimension, while retaining the same component-level equations and parameterization. The benchmarks span 53 differential states in the IEEE 9-bus system to 558 differential states in the IEEE 118-bus system and combine machine electromechanical dynamics, subtransient electrical dynamics, excitation and governor controls, fast grid-following inverter control loops, nonlinear converter current limits, induction-motor dynamics, voltage-dependent loads, algebraic network coupling, and discrete disturbance events within one common RMS transient-simulation framework.

Figure 2 illustrates a representative transient response for the IEEE 118-bus system, the largest network considered. RAPTOR closely follows the high-accuracy reference across machine, voltage, motor, and inverter states under the three disturbance events while requiring only 32 accepted intervals, whose endpoints are indicated by the red markers.

Table II quantifies the computational mechanism illustrated by Fig. 2. Across the five networked benchmarks, RAPTOR requires only 22–32 accepted intervals, compared with 192–234 advances for BDF, 123–153 for Radau, and 226–342 for LSODA. Despite covering substantially more physical time per interval, the RAPTOR nonlinear solve requires only 1.57–2.11 modified-Newton coefficient corrections per accepted interval. Jacobian reuse further limits the number of Jacobian constructions to 1–5 over the complete simulation. The factorization reduction is particularly pronounced relative to Radau, reaching  $396/32 = 12.4\times$  on IEEE 118. Relative to BDF, the



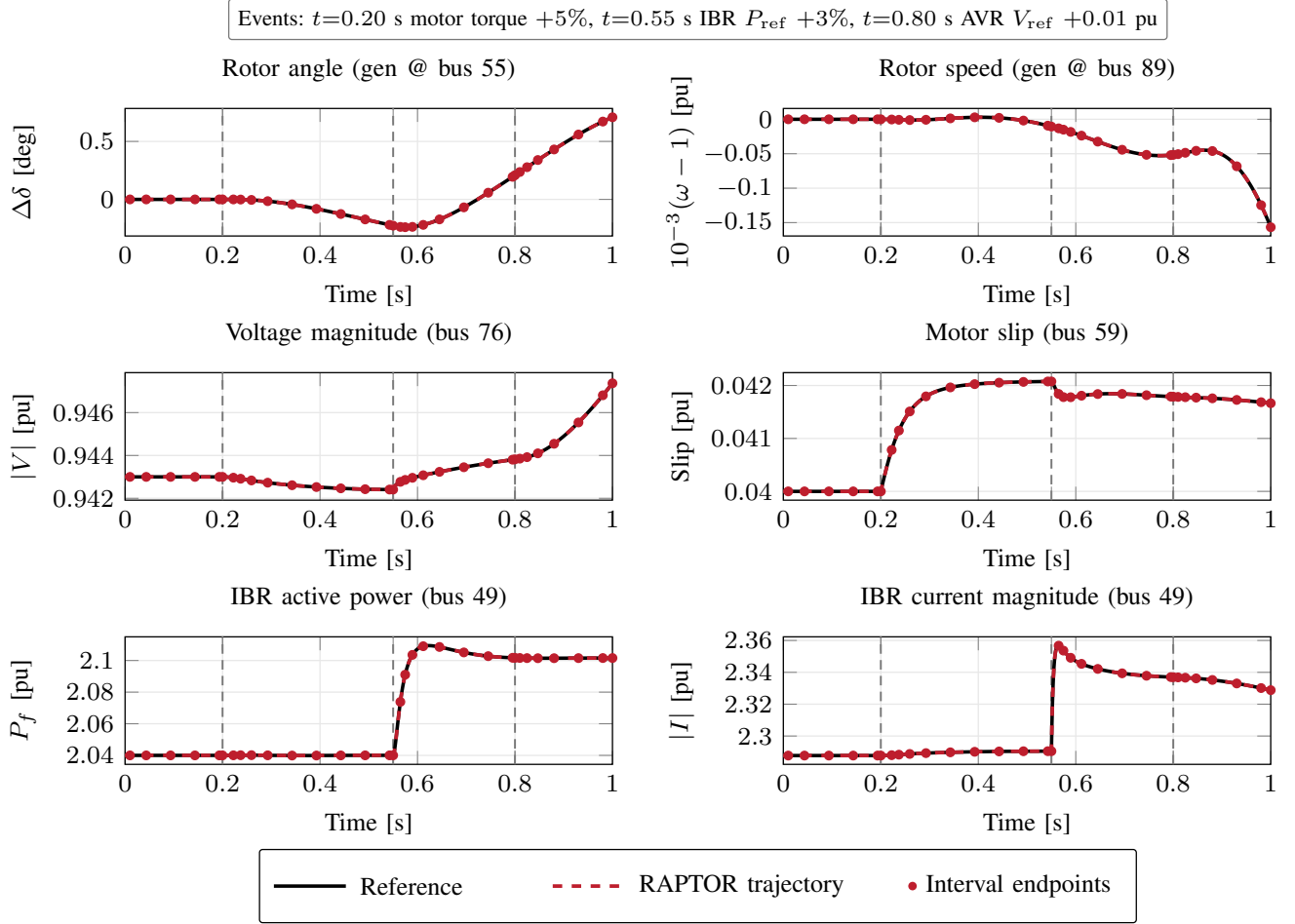


Fig. 2: IEEE 118-bus transient response: high-accuracy reference (solid black) and RAPTOR (dashed red). Red bullets mark the endpoints of the 32 accepted RAPTOR intervals. Vertical dashed lines indicate the motor-torque, inverter active-power-reference, and AVR-reference events.

TABLE II: Sequential advancement and nonlinear-solve effort for the selected accuracy-qualified IEEE RMS benchmark configurations. “Corr.” denotes RAPTOR modified-Newton (Chord) coefficient corrections. The LU columns report the measured number of matrix factorizations.

| System   | Intervals | RAPTOR |            |      |    | BDF  |    | Radau |     | LSODA |
|----------|-----------|--------|------------|------|----|------|----|-------|-----|-------|
|          |           | Corr.  | Corr./int. | Jac. | LU | Adv. | LU | Adv.  | LU  |       |
| IEEE 9   | 28        | 44     | 1.57       | 3    | 64 | 193  | 63 | 123   | 286 | 246   |
| IEEE 14  | 22        | 37     | 1.68       | 4    | 68 | 192  | 62 | 126   | 356 | 233   |
| IEEE 39  | 27        | 57     | 2.11       | 4    | 88 | 234  | 72 | 145   | 334 | 342   |
| IEEE 57  | 23        | 43     | 1.87       | 5    | 80 | 199  | 68 | 150   | 378 | 258   |
| IEEE 118 | 32        | 51     | 1.59       | 1    | 32 | 195  | 62 | 153   | 396 | 226   |

factorization counts are comparable on the smaller systems, showing that RAPTOR’s speedup there is primarily associated with fewer sequential advances and inexpensive convergence of the longer interval solves rather than fewer factorizations alone.

#### F. Benchmark Parameters and Reproducibility

This appendix collects the benchmark-specific physical parameters, controller gains, initialization settings, and internal RAPTOR numerical parameters used in the numerical experiments.

#### G. Standalone RMS Grid-Following Inverter

The standalone RMS inverter uses a nominal frequency of 50 Hz,  $\omega_N = 2\pi 50$  rad/s, and the parameter values in Table III.

The nominal initial state is

$$\mathbf{x}_0 = [0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1]^T.$$

The additional initial conditions perturb  $i_d$  by  $+0.2$  pu,  $i_q$  by  $+0.1$  pu,  $P_{\text{filt}}$  to 0.8 pu,  $V_{\text{filt}}$  to 1.05 pu, and  $\theta_{\text{pll}}$  by  $+0.05$  rad.

For the weak-grid sensitivity experiment, the controller parameters remain unchanged and the network impedance is

TABLE III: Standalone RMS grid-following inverter parameters.

| Parameter              | Description                         | Value               |
|------------------------|-------------------------------------|---------------------|
| $P_{\text{ref}}$       | Active-power reference              | 1.0 pu              |
| $Q_{\text{ref}}$       | Reactive-power reference            | 0.0 pu              |
| $V_{\text{ref}}$       | Voltage reference                   | 1.0 pu              |
| $\omega_{\text{ref}}$  | Frequency reference                 | $2\pi 50$ rad/s     |
| $v_{\text{dc}}$        | DC-link voltage                     | 1.0 pu              |
| $k_{pdp}$              | Active-power proportional gain      | 1.0                 |
| $k_{idp}$              | Active-power integral gain          | 50                  |
| $k_{pdq}$              | Reactive-power proportional gain    | 1.0                 |
| $k_{idq}$              | Reactive-power integral gain        | 50                  |
| $k_{pid}$              | $d$ -axis current proportional gain | 1.0                 |
| $k_{iid}$              | $d$ -axis current integral gain     | 500                 |
| $k_{piq}$              | $q$ -axis current proportional gain | 1.0                 |
| $k_{iiq}$              | $q$ -axis current integral gain     | 500                 |
| $k_{ppl}$              | PLL proportional gain               | 30                  |
| $k_{ipl}$              | PLL integral gain                   | 900                 |
| $R$                    | Series resistance                   | 0.01 pu             |
| $L$                    | Series inductance                   | $0.1/(2\pi 50)$     |
| $\omega_{\text{filt}}$ | Measurement-filter rate             | $25 \text{ s}^{-1}$ |

TABLE IV: Internal RAPTOR settings for the standalone RMS inverter.

| Parameter               | Meaning                              | Value                        |
|-------------------------|--------------------------------------|------------------------------|
| $M$                     | Collocation points                   | 8                            |
| $\Delta t_{\text{min}}$ | Minimum interval length              | $5 \times 10^{-5} \text{ s}$ |
| $s_0$                   | Initial-interval safety multiplier   | 1.0                          |
| $\nu_{\text{max}}$      | Maximum rejected retries             | 5                            |
| $k_P$                   | PI proportional exponent coefficient | 0.7                          |
| $k_I$                   | PI integral exponent coefficient     | 0.3                          |
| $\gamma_{\text{min}}$   | Minimum interval factor              | 0.2                          |
| $\eta_{\text{saf}}$     | PI safety factor                     | 0.9                          |
| $\gamma_{\text{max}}$   | Maximum interval factor              | 2.0                          |
| $\lambda_{\text{cont}}$ | Continuation regularization          | $10^{-3}$                    |
| Chord rtol              | Nonlinear-solver relative tolerance  | $10^{-3}$                    |
| Chord atol              | Nonlinear-solver absolute tolerance  | $10^{-3}$                    |
| Max. Chord steps        | Nonlinear iterations per attempt     | 300                          |
| Random seed             | RBF sampling seed                    | 0                            |

varied with SCR according to

$$L(\text{SCR}) = \frac{1}{2\pi 50 \text{ SCR}}, \quad R(\text{SCR}) = \frac{0.05}{\text{SCR}},$$

for

$$\text{SCR} \in \{5, 3, 2, 1.2\}.$$

The common internal RAPTOR settings used for this model are summarized in Table IV.

The PI interval controller uses

$$\Delta t_{k+1} = 0.9 \Delta t_k \text{ clip} \left[ \text{err}_k^{-0.7/r} \text{err}_{k-1}^{-0.3/r}, 0.2, 2.0 \right], \quad r = \nu + 1$$

#### H. Cascaded M-Section Transmission Line

The three-section line uses the physical parameters in Table V.

The resulting section parameters are

$$R' = \frac{R}{2} = 0.0025, \quad L' = \frac{L}{2} = \frac{0.05}{2\pi 50},$$

$$C_1 = C_3 = \frac{1-N}{2}C = 3.3 \times 10^{-6}, \quad C_2 = NC = 6.5934 \times 10^{-6},$$

and

$$G_1 = G_3 = \frac{1-N}{2}G = 10^{-4}, \quad G_2 = NG = 0.1998.$$

TABLE V: Cascaded M-section line parameters.

| Parameter       | Description                      | Value           |
|-----------------|----------------------------------|-----------------|
| $R$             | Total series resistance          | 0.005           |
| $L$             | Total series inductance          | $0.1/(2\pi 50)$ |
| $C$             | Total shunt capacitance          | 0.0066          |
| $G$             | Total shunt conductance          | 0.2             |
| $\omega$        | Synchronous angular frequency    | $2\pi 50$ rad/s |
| $N$             | Central-section partition factor | 0.999           |
| $I_{\text{in}}$ | Boundary-input amplitude         | 0.02            |
| $f_{\text{in}}$ | Boundary-input frequency         | 50 Hz           |

TABLE VI: Internal RAPTOR settings for the M-section benchmark.

| Parameter               | Meaning                              | Value                        |
|-------------------------|--------------------------------------|------------------------------|
| $M$                     | Collocation points                   | 10                           |
| $N$                     | Gaussian RBFs in tuned configuration | 10                           |
| $\Delta t_{\text{min}}$ | Minimum interval                     | $5 \times 10^{-5} \text{ s}$ |
| $s_0$                   | Initial-interval safety multiplier   | 1.0                          |
| $\nu_{\text{max}}$      | Maximum retries                      | 8                            |
| $k_P$                   | PI proportional coefficient          | 0.7                          |
| $k_I$                   | PI integral coefficient              | 0.3                          |
| $\gamma_{\text{min}}$   | Minimum interval factor              | 0.2                          |
| $\eta_{\text{saf}}$     | PI safety factor                     | 0.9                          |
| $\gamma_{\text{max}}$   | Maximum interval factor              | 2.0                          |
| Rejection factor        | Maximum rejected-interval reduction  | 0.5                          |
| Ridge parameter         | Linear collocation regularization    | $10^{-10}$                   |
| Random seed             | Initial-condition seed               | 7                            |
| Timing repetitions      | Repeated executions                  | 3                            |

The unperturbed initial state is  $\mathbf{x}_0 = \mathbf{0}$ . Randomized initial conditions are formed by independent zero-mean Gaussian perturbations with standard deviation 0.04 for both voltage and current states. The initial-condition random seed is 7.

#### I. Networked IEEE RMS DAE Benchmarks

The IEEE 9-, 14-, 39-, 57-, and 118-bus systems use a 100-MVA system base and 60-Hz nominal frequency. The dynamic-device composition is given in Table VII.

The target inverter share is

$$\eta_{\text{IBR}} = 0.20,$$

and the target induction-motor fraction at a selected dynamic load bus is

$$\eta_{\text{IM}} = 0.25.$$

At most six motor buses are used. Candidate motor buses satisfy

$$P_L > 0.03 \text{ pu}, \quad Q_L > 0.005 \text{ pu}.$$

The fractional IBR assignments at marginal plants are

| System       | 9      | 14     | 39     | 57     | 118     |
|--------------|--------|--------|--------|--------|---------|
| Marginal bus | 1      | 1      | 37     | 12     | 25      |
| IBR fraction | 0.8893 | 0.0623 | 0.9288 | 0.6959 | 0.5313. |

Machine-to-machine diversity is introduced deterministically using

$$H_r = 3.5 + 0.4[(r \bmod 4) + 1] \text{ s},$$

$$T'_{do,r} = 6.0 + 0.7(r \bmod 4) \text{ s}, \quad x''_r = 0.20 + 0.01(r \bmod 3),$$

giving

$$H_r \in \{3.9, 4.3, 4.7, 5.1\} \text{ s},$$

$$T'_{do,r} \in \{6.0, 6.7, 7.4, 8.1\} \text{ s}, \quad x''_r \in \{0.20, 0.21, 0.22\}.$$

TABLE VII: Composition of the networked IEEE RMS benchmarks.

| System   | Buses | Branches | SG | IBR | IM | $n_x$ | $P_{IM}/P_L$ | IBR buses                          | Motor buses       |
|----------|-------|----------|----|-----|----|-------|--------------|------------------------------------|-------------------|
| IEEE 9   | 9     | 9        | 3  | 1   | 3  | 53    | 25.00%       | 1                                  | 5,7,9             |
| IEEE 14  | 14    | 20       | 4  | 2   | 6  | 86    | 17.86%       | 1,2                                | 2,3,6,9,13,14     |
| IEEE 39  | 39    | 46       | 8  | 3   | 6  | 134   | 11.72%       | 30,34,37                           | 3,4,8,16,20,39    |
| IEEE 57  | 57    | 80       | 6  | 2   | 6  | 106   | 9.49%        | 3,12                               | 1,6,8,9,12,16     |
| IEEE 118 | 118   | 186      | 44 | 11  | 6  | 558   | 5.12%        | 12,25,31,46,49,54,59,61,87,103,111 | 15,42,54,59,80,90 |

TABLE VIII: Synchronous-machine, AVR, and governor parameters.

| Parameter  | Description                          | Value   |
|------------|--------------------------------------|---------|
| $x_d$      | $d$ -axis synchronous reactance      | 1.80    |
| $x_q$      | $q$ -axis synchronous reactance      | 1.70    |
| $x'_d$     | $d$ -axis transient reactance        | 0.30    |
| $x'_q$     | $q$ -axis transient reactance        | 0.55    |
| $T'_{d0}$  | $q$ -axis transient time constant    | 0.40 s  |
| $T'_{d0}'$ | $d$ -axis subtransient time constant | 0.030 s |
| $T'_{q0}$  | $q$ -axis subtransient time constant | 0.050 s |
| $D$        | Damping coefficient                  | 1.0     |
| $K_A$      | AVR gain                             | 20      |
| $T_A$      | AVR time constant                    | 0.050 s |
| $T_r$      | Voltage-measurement time constant    | 0.020 s |
| $R_g$      | Governor droop                       | 0.05    |
| $T_g$      | Governor time constant               | 0.20 s  |
| $T_t$      | Turbine time constant                | 0.50 s  |

TABLE IX: Grid-following IBR parameters for the IEEE RMS benchmarks.

| Parameter    | Description                          | Value   |
|--------------|--------------------------------------|---------|
| $k_{p,pll}$  | PLL proportional gain                | 35      |
| $k_{i,pll}$  | PLL integral gain                    | 600     |
| $T_{pq}$     | Power-filter time constant           | 0.020 s |
| $k_{p,P}$    | Active-power proportional gain       | 0.40    |
| $k_{i,P}$    | Active-power integral gain           | 15      |
| $k_{p,Q}$    | Reactive-power proportional gain     | 0.40    |
| $k_{i,Q}$    | Reactive-power integral gain         | 15      |
| $T_i$        | Inner current-response time constant | 0.004 s |
| $V_\epsilon$ | Voltage regularization floor         | 0.20 pu |

The inverter current limit is initialized as

$$I_{\max,i} = 1.30 \max(|I_{dq,i}(0)|, 0.10).$$

The internal RAPTOR parameters used in the IEEE configuration sweep are summarized in Table XI.

The tested Jacobian-reuse periods are

$$n_{\text{reuse}} \in \{8, 12\}$$

accepted intervals, and both one-point and two-point coefficient initialization are tested. Gaussian centers are

$$\beta_j = \frac{j + \frac{1}{2}}{N}, \quad j = 0, \dots, N-1,$$

and their positive shape parameters are initialized using random seed 0.

### J. EMT Grid-Following Inverter

The production EMT grid-following inverter uses a 50-Hz base frequency, with the electrical quantities in the source parameter file converted to the continuous-time model according to

$$\omega_N = 2\pi 50.$$

TABLE X: Induction-motor parameters for the IEEE RMS benchmarks.

| Parameter        | Description                | Value   |
|------------------|----------------------------|---------|
| $r_s$            | Stator resistance          | 0.020   |
| $L_s$            | Synchronous inductance     | 2.50    |
| $L'$             | Transient inductance       | 0.18    |
| $L''$            | Subtransient inductance    | 0.12    |
| $T'$             | Transient time constant    | 0.15 s  |
| $T''$            | Subtransient time constant | 0.025 s |
| $H_m$            | Motor inertia constant     | 0.50 s  |
| $E_{\text{trq}}$ | Mechanical torque exponent | 2.0     |
| $s_m(0)$         | Initial slip               | 0.04    |
| $\omega_m(0)$    | Initial motor speed        | 0.96 pu |

TABLE XI: Internal RAPTOR parameters for the IEEE RMS DAE experiments.

| Parameter             | Meaning                          | Value                |
|-----------------------|----------------------------------|----------------------|
| $N$                   | Gaussian RBFs per state          | 7                    |
| $M$                   | Collocation points               | 7                    |
| $N_{\text{audit}}$    | Off-grid audit points            | 9                    |
| $\Delta t_0$          | Initial interval                 | $10^{-2}$ s          |
| $\Delta t_{\min}$     | Minimum interval                 | $2 \times 10^{-5}$ s |
| $\text{atol}_R$       | Fixed residual absolute scale    | $2 \times 10^{-7}$   |
| Newton tolerance      | Coefficient-correction tolerance | $3 \times 10^{-2}$   |
| Maximum corrections   | Chord/Newton safety cap          | 5                    |
| Jacobian refresh      | Internal refresh parameter       | 2                    |
| GMRES rtol            | Iterative linear-solve tolerance | $10^{-4}$            |
| $\eta_{\text{saf}}$   | Interval safety factor           | 0.93                 |
| $\gamma_{\min}$       | Minimum growth factor            | 0.35                 |
| $\gamma_{\max}$       | Maximum growth factor            | 2.5                  |
| $\gamma_{\text{rej}}$ | Rejected-interval factor         | 0.50                 |
| Random seed           | Gaussian-width seed              | 0                    |

Numerically,

$$L_c = L_g = 3.18310 \times 10^{-4}, \quad C_f = 3.76996 \times 10^{-6}.$$

The initialization uses

$$|V_{\text{PCC}}(0)| = 1, \quad \angle V_{\text{PCC}}(0) = 0,$$

$$|I_g(0)| = 0, \quad \angle I_g(0) = 0,$$

with initial event states

$$s_e(0) = 1, \quad P^*(0) = 0, \quad Q^*(0) = 0.$$

For the voltage-sag trajectory used in the reported waveform comparison,

$$s_e = \begin{cases} 0.8, & 1.1 \leq t < 1.2 \text{ s}, \\ 1.0, & \text{otherwise}, \end{cases}$$

and a forced event-localization interval of  $10^{-5}$  s is used immediately around the event.

TABLE XII: EMT grid-following inverter electrical and controller parameters.

| Parameter   | Description                          | Value                       |
|-------------|--------------------------------------|-----------------------------|
| $L_g$       | Grid-side inductance                 | $0.1/(2\pi 50)$             |
| $R_{lg}$    | Grid-side resistance                 | 0.005                       |
| $L_c$       | Converter-side inductance            | $0.1/(2\pi 50)$             |
| $R_c$       | Converter-side resistance            | 0.005                       |
| $C_f$       | Filter capacitance                   | $0.0011840835645/(2\pi 50)$ |
| $R_f$       | Damping resistance                   | 0.0757                      |
| $K_p^{cc}$  | Current-controller proportional gain | 0.1                         |
| $K_i^{cc}$  | Current-controller integral gain     | 20                          |
| $K_p^{pc}$  | Power-controller proportional gain   | 0.5                         |
| $K_i^{pc}$  | Power-controller integral gain       | 30                          |
| $K_p^{pll}$ | PLL proportional gain                | 25                          |
| $K_i^{pll}$ | PLL integral gain                    | 300                         |

TABLE XIII: Internal RAPTOR parameters for the EMT grid-following inverter.

| Parameter                  | Meaning                              | Value       |
|----------------------------|--------------------------------------|-------------|
| $N$                        | Gaussian RBFs per differential state | 6           |
| $M$                        | Collocation points                   | 6           |
| $\lambda_{alg}$            | Algebraic-residual weight            | 1.0         |
| $\Delta t_{min}$           | Minimum interval                     | $10^{-3}$ s |
| $s_0$                      | Initial-interval safety multiplier   | 1.0         |
| $\nu_{max}$                | Maximum retry count                  | 20          |
| $k_P$                      | PI proportional coefficient          | 0.7         |
| $k_I$                      | PI integral coefficient              | 0.3         |
| $\gamma_{min}$             | Minimum interval factor              | 0.2         |
| $\eta_{saf}$               | PI safety factor                     | 0.9         |
| $\gamma_{max}$             | Maximum interval factor              | 2.0         |
| $\epsilon_{PI}$            | PI error floor                       | $10^{-8}$   |
| $\tau_{relax}$             | Time-scale limiter factor            | 1.0         |
| Random seed                | RBF sampling seed                    | 0           |
| Maximum Chord steps        | Nonlinear solve cap                  | 40          |
| Algebraic projection steps | Post-event projection cap            | 15          |
| Chord rtol                 | Interval nonlinear solve             | $10^{-8}$   |
| Chord atol                 | Interval nonlinear solve             | $10^{-8}$   |
| Projection rtol            | Algebraic projection                 | $10^{-10}$  |
| Projection atol            | Algebraic projection                 | $10^{-10}$  |

The interval controller uses the endpoint differential residual,

$$\text{err}_k = \left[ \frac{1}{n_x} \sum_i \left( \frac{r_{k,i}}{\pi_{atol} + \pi_{rtol}|x_{k,i}|} \right)^2 \right]^{1/2},$$

with  $k_P = 0.7$  and  $k_I = 0.3$ , and proposes

$$\Delta t_{k+1} = 0.9 \Delta t_k \text{ clip } \left[ \text{err}_k^{-0.7/r} \text{err}_{k-1}^{-0.3/r}, 0.2, 2.0 \right], \quad \text{at } r = \nu + 1.$$

### K. Synchronous-Machine EMT Benchmark

The synchronous-machine EMT benchmark uses the operating point

$$P_0 = 0.5 \text{ pu}, \quad Q_0 = 0.1 \text{ pu}, \quad V_0 = 1.0 \text{ pu},$$

and a 50-Hz nominal frequency. Its primitive and derived parameters are listed in Table XIV.

The infinite-bus phasor resulting from the initialized operating point is

$$|E_\infty| = 0.9800133, \quad \angle E_\infty = -0.101192 \text{ rad},$$

and the initialized rotor electrical angle is

$$\theta_0 = 0.767791 \text{ rad}.$$

TABLE XIV: Synchronous-machine EMT benchmark parameters.

| Parameter  | Description                          | Value                    |
|------------|--------------------------------------|--------------------------|
| $X_e$      | External line reactance              | 0.20                     |
| $R_e$      | External line resistance             | 0.01                     |
| $L_e$      | External line inductive parameter    | 0.20                     |
| $R_a$      | Stator resistance                    | 0.005                    |
| $X_d$      | $d$ -axis reactance                  | 2.4                      |
| $X_q$      | $q$ -axis reactance                  | 2.4                      |
| $X_l$      | Leakage reactance                    | 0.2                      |
| $X'_d$     | $d$ -axis transient reactance        | 0.40                     |
| $X'_q$     | $q$ -axis transient reactance        | 0.25                     |
| $L_{qo}$   | Zero-sequence inductance             | 0.1                      |
| $T'_{do}$  | Open-circuit $d$ -axis time constant | 7.0 s                    |
| $T'_{qo}$  | Open-circuit $q$ -axis time constant | 0.3 s                    |
| $L_{dd}$   | $d$ -axis self inductance            | 2.4                      |
| $L_{df}$   | $d$ -axis mutual inductance          | 2.2                      |
| $L_{ff}$   | Field self inductance                | 2.42                     |
| $R_f$      | Field resistance                     | $1.10044 \times 10^{-3}$ |
| $L_{qq}$   | $q$ -axis self inductance            | 2.4                      |
| $L_{qq1}$  | $q$ -axis mutual inductance          | 2.2                      |
| $L_{q1q1}$ | Damper self inductance               | 2.25116                  |
| $R_{q1}$   | Damper resistance                    | $2.38856 \times 10^{-2}$ |

TABLE XV: Internal RAPTOR parameters for the synchronous-machine EMT benchmark.

| Parameter                   | Meaning                              | Value                |
|-----------------------------|--------------------------------------|----------------------|
| $N$                         | Gaussian RBFs per differential state | 6                    |
| $M$                         | Collocation points                   | 6                    |
| $\lambda_{alg}$             | Algebraic-residual weight            | 10                   |
| $\Delta t_{min}$            | Minimum interval                     | $5 \times 10^{-4}$ s |
| $s_0$                       | Initial-interval multiplier          | $10^{-3}$            |
| $\nu_{max}$                 | Maximum retries                      | 10                   |
| $k_P$                       | PI proportional coefficient          | 0.7                  |
| $k_I$                       | PI integral coefficient              | 0.3                  |
| $\gamma_{min}$              | Minimum interval factor              | 0.2                  |
| $\eta_{saf}$                | PI safety factor                     | 0.9                  |
| $\gamma_{max}$              | Maximum interval factor              | 2.1                  |
| $\epsilon_{PI}$             | PI error floor                       | $10^{-8}$            |
| $\tau_{relax}$              | Time-scale limiter                   | 1.0                  |
| Random seed                 | RBF seed                             | 0                    |
| Maximum Chord steps         | Nonlinear solve cap                  | 20                   |
| Algebraic projection steps  | Projection cap                       | 10                   |
| Event localization interval | Post-event interval                  | $10^{-6}$ s          |

The switching event sets the infinite-bus source multiplier from

$$s_e = 1 \quad \longrightarrow \quad s_e = 0$$

$$t_e = 0.05 \text{ s},$$

and the post-event localization interval is  $10^{-6}$  s.

For the Gaussian RBF representation in this EMT benchmark, the centers are uniformly distributed on  $[0, 1]$ , and the sampled shape parameters use random seed 0. The adaptive controller again uses

$$k_P = 0.7, \quad k_I = 0.3,$$

with acceptance whenever the scaled endpoint residual satisfies

$$\text{err}_k < 1.$$

### L. Comparison of Temporal Representations and Classical Integrators

This appendix provides numerical experiments aimed at isolating the effect of the temporal representation used within

RAPTOR and comparing the resulting solver with established implicit time-integration methods. These experiments complement the main benchmark results by distinguishing between (i) the approximation efficiency of Gaussian radial basis functions (RBFs), Chebyshev–Gauss–Lobatto (CGL) polynomials, and Legendre–Gauss–Lobatto (LGL) polynomials, and (ii) the overall computational efficiency of RAPTOR relative to conventional methods such as Radau and BDF.

For the representation comparisons, the RBF, CGL, and LGL formulations use comparable temporal approximation sizes and collocation budgets. The polynomial formulations retain their native Lobatto node distributions. All RAPTOR variants employ the same physical model, event schedule, adaptive acceptance criterion, off-grid trajectory audit, and error metrics. Consequently, differences between these experiments primarily reflect the effectiveness of the temporal basis rather than differences in the physical problem or accuracy requirement.

1) *Comparison setup and reproducibility:* The four comparisons use different physical horizons because they target different numerical behaviours. The cascaded M-section line is simulated for 2 ms, which contains several cycles of its approximately 4.9–7.0 kHz electrical modes. A balanced 50 Hz current injection with amplitude 0.50 pu is applied at the initial time, and the magnitude of the first branch current is additionally monitored. The weak-grid RMS inverter is simulated for 50 ms at SCR = 1.2, after its active-power reference changes from 1.0 to 1.10 pu. This horizon contains the rapid current-control and PLL response together with the slower filter and outer-control evolution.

The grid-following EMT inverter is simulated for 1 s. The source-voltage magnitude decreases to 0.65 pu at  $t = 0.20$  s and is restored at  $t = 0.22$  s, while the active-power reference changes from 0.8 to 0.9 pu at  $t = 0.60$  s. The EMT synchronous-generator comparison uses a 0.1 s horizon and contains the switching event at  $t = 0.05$  s.

The M-section comparison uses

$$N = 0.999, \quad R_0 = 0.005, \quad L_0 = \frac{0.1}{2\pi 50}, \quad C = 0.0066$$

The weak-grid RMS comparison uses the inverter equations given earlier with

$$K_{p,PLL} = 30, \quad K_{i,PLL} = 900, \quad K_{p,cc} = 1, \quad K_{i,cc} = 500, \\ K_{p,PQ} = 1, \quad K_{i,PQ} = 50,$$

and a measurement-filter rate of  $25 \text{ s}^{-1}$ .

The 1 s EMT comparison uses the disclosed representative parameter set

$$R_c = R_{lg} = 0.01, \quad L_c = L_g = \frac{0.1}{2\pi 50}, \quad R_f = 1, \quad C_f = \frac{0.05}{2\pi 50}, \\ K_{p,PQ} = 1, \quad K_{i,PQ} = 50, \quad K_{p,cc} = 1, \quad K_{i,cc} = 500, \\ K_{p,PLL} = 30, \quad K_{i,PLL} = 900.$$

This parameter set belongs specifically to the 1 s numerical comparison reported in this appendix.

a) *Reference calculations and error measures.:* The M-section line reference is obtained from the matrix exponential of the augmented linear system at 4001 common output times and is cross-checked with Radau using  $\text{rtol} = 2 \times 10^{-11}$  and  $\text{atol} = 2 \times 10^{-13}$ . The RMS reference uses the same tight Radau tolerances and is independently cross-checked with tight BDF. The 1 s EMT reference uses event-aligned high-accuracy Radau and is independently cross-checked with event-aligned DOP853. The EMT synchronous-generator comparison uses the refined fully coupled trapezoidal reference described above, with the  $10^{-7}$  s calculation used for the reported representation errors.

For differential state  $j$ , define the reference scale

$$s_j = \max \left\{ \text{range}(x_{j,\text{ref}}), 0.05 \max_t |x_{j,\text{ref}}(t)|, s_{j,\text{floor}} \right\}. \quad (193)$$

The scaled trajectory error is

$$E_x = \frac{1}{n_x} \sum_{j=1}^{n_x} \left[ \frac{1}{N_t} \sum_{m=1}^{N_t} \left( \frac{x_j(t_m) - x_{j,\text{ref}}(t_m)}{s_j} \right)^2 \right]^{1/2}. \quad (194)$$

The same construction defines  $E_y$  for recovered algebraic variables. For the EMT inverter,  $E_{x,\text{event}}$  and  $E_{y,\text{event}}$  use the output points in the windows

$$[t_e - 2 \text{ ms}, t_e + 30 \text{ ms}], \quad t_e \in \{0.20, 0.22, 0.60\} \text{ s}.$$

The quantity  $E_I$  is the scaled grid-current waveform error, while

$$|I_{\max} - I_{\max}^{\text{ref}}|$$

measures the error in the maximum grid-current magnitude.

b) *Comparison requirements.:* For the M-section line, a configuration qualifies when

$$E_x \leq 10^{-4}, \quad |I_{\text{br},\max} - I_{\text{br},\max}^{\text{ref}}| \leq 5 \times 10^{-5} \text{ pu}.$$

The comparison then selects the lowest-runtime configuration satisfying both requirements.

The weak-grid RMS experiment is an equal-size representation comparison rather than a target-matched solver comparison. The Gaussian-RBF, CGL, and LGL representations all use approximation order  $p = 8$  and  $K = 16$  time elements, and their resulting runtime and trajectory error are compared directly.

For the 1 s EMT inverter, the accuracy-oriented comparison requires

$$E_x \leq 3 \times 10^{-5}, \quad E_{x,\text{event}} \leq 1.5 \times 10^{-4}, \quad (195)$$

$$E_I \leq 6 \times 10^{-5}, \quad |I_{\max} - I_{\max}^{\text{ref}}| \leq 10^{-5} \text{ pu}, \quad (196)$$

$$E_y \leq 10^{-3}, \quad E_{y,\text{event}} \leq 3 \times 10^{-3}. \quad (197)$$

These are numerical comparison requirements used to select comparable solver configurations; they are not protection, equipment, or grid-code limits.

c) *Selected 1 s EMT configurations.:* The conventional-solver tolerances were audited before the final comparison. For Radau, all tested relative tolerances from  $2 \times 10^{-3}$  through  $1.05 \times 10^{-3}$  fail at least one accuracy requirement, while  $10^{-3}$  satisfies all six. For BDF, tested tolerances from  $3 \times 10^{-5}$

TABLE XVI: Numerical settings used in the 1 s EMT inverter comparison.

| Method      | Setting                                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAPTOR      | Degree 10; 21 independent audit points; residual tolerance $3 \times 10^{-3}$ ; initial interval 0.2 ms; maximum interval 50 ms; projected continuation; at most three Chord updates before a Jacobian refresh. |
| CGL         | Degree 6; 15 independent audit points; residual tolerance $5 \times 10^{-3}$ ; event-aligned adaptive element length.                                                                                           |
| LGL         | Degree 6; 15 independent audit points; residual tolerance $4 \times 10^{-3}$ for the accuracy-oriented configuration.                                                                                           |
| Radau IIA   | Analytic Jacobian; event-aligned restart; $\text{rtol} = 10^{-3}$ .                                                                                                                                             |
| BDF         | Analytic Jacobian; event-aligned restart; $\text{rtol} = 7 \times 10^{-6}$ .                                                                                                                                    |
| Trapezoidal | Newton solve with analytic Jacobian; step-doubling controller; $\text{rtol} = 10^{-4}$ ; exact event alignment.                                                                                                 |

through  $10^{-5}$  fail at least one requirement, while  $7 \times 10^{-6}$  satisfies all six. The selected conventional configurations are therefore the loosest tested settings satisfying the complete accuracy requirement.

d) *EMT synchronous-generator equal-size settings.*: The adaptive equal-size comparison uses the same approximation order, collocation count, independent audit count, residual tolerances, and interval controller for the Gaussian-RBF, CGL, and LGL representations. Specifically,

$$p = 8, \quad N_{\text{coll}} = 8, \quad N_{\text{audit}} = 16.$$

The initial, minimum, and maximum interval lengths are

$$\Delta t_0 = 0.2 \text{ ms}, \quad \Delta t_{\min} = 0.02 \text{ ms}, \quad \Delta t_{\max} = 8 \text{ ms}.$$

The residual relative and absolute tolerances are  $10^{-3}$  and  $10^{-8}$ , respectively. The adaptive controller uses acceptance threshold 1, safety factor 0.86, minimum and maximum interval factors 0.25 and 2.0, and coefficients

$$k_P = 0.16, \quad k_I = 0.04.$$

At most 10 rejected retries and 8 Chord updates are permitted, with Jacobian refresh after three updates when required. The Gaussian-RBF representation uses width scale 2.0. The model-specific affine shortcut is disabled so that all three representations are evaluated using the same generic adaptive implementation.

e) *Timing protocol.*: All reported timings use double precision and one CPU thread. The numerical environment uses Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3, and JAX 0.9.0.1. The environment variables `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, `MKL_NUM_THREADS`, and `NUMEXPR_NUM_THREADS` are set to one.

One warm-up run is performed before timing the local trajectory methods so that compilation is excluded. For the short M-section and RMS experiments, the local numerical system is assembled once and reused for all timing repetitions of a selected configuration. The same policy is applied to the Gaussian-RBF, CGL, and LGL representations.

For the final RAPTOR–Radau accuracy-oriented EMT comparison, the methods are run in nine interleaved pairs. RAPTOR is faster in all nine pairs; the Radau-to-RAPTOR runtime ratio ranges from 1.16 to 1.29, with median 1.20. The CGL, LGL, and BDF values use five timing repetitions and report the

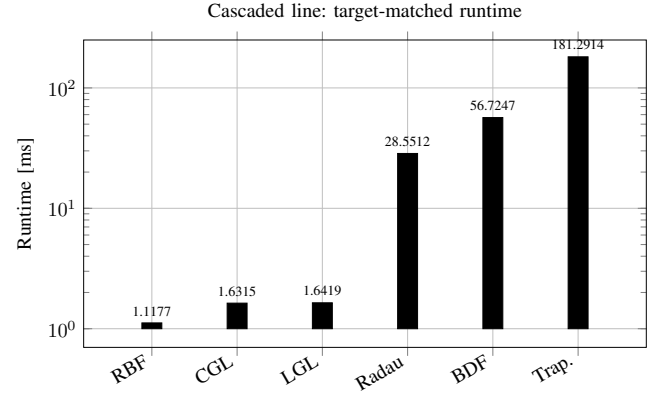


Fig. 3: Target-matched runtime comparison for the cascaded transmission-line benchmark. RAPTOR reaches the prescribed trajectory-accuracy target with the lowest runtime in this experiment.

median runtime. Accepted RAPTOR intervals, CGL or LGL spectral elements, and conventional timesteps have different internal computational costs. Their accepted counts are therefore reported to characterize sequential numerical work, while runtime and trajectory accuracy are the primary end-to-end comparison measures.

2) *Cascaded transmission-line system.*: The cascaded transmission-line benchmark provides a target-matched comparison, where the methods are compared at a prescribed trajectory-accuracy level. Random-Projection Physics-Informed Transient Solver (RAPTOR) requires approximately 1.12 ms, whereas the CGL and LGL spectral variants require approximately 1.63 ms and 1.64 ms, respectively. The corresponding Radau and BDF runtimes are approximately 28.6 ms and 56.7 ms.

RAPTOR therefore achieves the lowest runtime at the prescribed accuracy level. The CGL approximation attains a somewhat smaller trajectory error, so the advantage observed here should be interpreted as a reduction in *time-to-target accuracy*, rather than as a claim that the Gaussian representation always minimizes the numerical error.

The RBF formulation reaches the required accuracy using only 12 accepted advances, compared with 24 for the CGL and LGL variants. This indicates that the localized Gaussian representation provides an efficient approximation of the multiple temporal scales present in the distributed line model.

3) *RMS inverter: equal-size representation comparison.*: To isolate the effect of the temporal basis, an equal-size comparison is performed for the RMS inverter model. The Gaussian-RBF, CGL, and LGL formulations use the same approximation order,  $p = 8$ , and the same number of intervals,  $K = 16$ .

The measured results are

$$\text{RBF: } 58.6 \text{ ms}, \quad \text{RMSE} = 3.52 \times 10^{-5}, \quad (198)$$

$$\text{CGL: } 74.7 \text{ ms}, \quad \text{RMSE} = 4.10 \times 10^{-5}, \quad (199)$$

$$\text{LGL: } 71.7 \text{ ms}, \quad \text{RMSE} = 6.29 \times 10^{-5}. \quad (200)$$

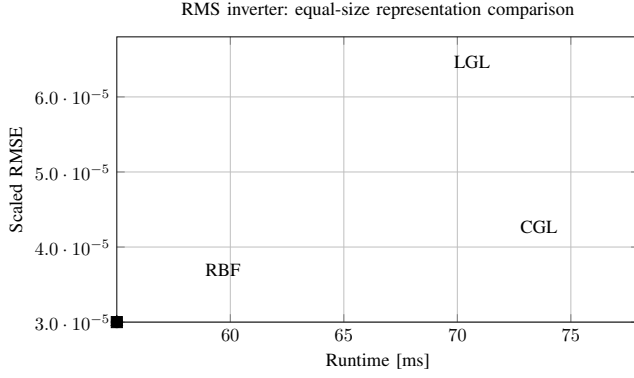


Fig. 4: Runtime–accuracy comparison for the equal-size Gaussian-RBF, CGL, and LGL temporal representations in the RMS inverter benchmark. All methods use the same temporal approximation size, with  $p = 8$  and  $K = 16$ .

Under the same representation budget, the Gaussian-RBF formulation is therefore both faster and more accurate than the two polynomial alternatives. Its runtime is approximately  $1.27\times$  lower than CGL and  $1.22\times$  lower than LGL.

This experiment provides a direct comparison of the approximation efficiency of the three temporal bases. In this RMS inverter model, the localized support provided by the overlapping Gaussian functions represents the network and controller dynamics efficiently without requiring additional temporal coefficients.

4) *EMT PLL inverter*: The EMT PLL inverter represents a more demanding nonlinear and event-driven case. The system combines converter-control dynamics, PLL dynamics, electrical network dynamics, and algebraic constraints, together with a sequence of disturbances.

For the EMT PLL experiment, RAPTOR requires approximately 1.10 s and accepts 256 temporal intervals. The corresponding runtimes for the other methods are approximately 1.63 s(CGL), 1.85 s(LGL), 1.33 s(Radau), 2.33 s(BDF).

The methods are assessed using simultaneous accuracy requirements on the differential states, event states, current waveform, peak current, algebraic variables, and algebraic variables near events. RAPTOR satisfies all of these criteria.

Among the methods satisfying the complete accuracy requirements, RAPTOR therefore provides the lowest runtime. Its runtime is approximately  $1.20\times$  lower than Radau,  $1.47\times$  lower than CGL,  $1.68\times$  lower than LGL, and  $2.11\times$  lower than BDF.

This benchmark illustrates a regime in which the localized nature of the Gaussian representation is particularly beneficial. The PLL, converter controllers, and electrical network introduce several interacting time scales, while the disturbances produce localized changes in the trajectory. Overlapping Gaussian functions can resolve these local temporal features without increasing the polynomial degree over the entire interval.

5) *EMT synchronous generator*: The EMT synchronous-machine benchmark represents a different computational regime. In the strict adaptive equal-size comparison, the mea-

sured results are approximately

$$\text{RBF: } 21.7 \text{ ms, } 49 \text{ accepted intervals, } (201)$$

$$\text{CGL: } 13.4 \text{ ms, } 31 \text{ accepted intervals, } (202)$$

$$\text{LGL: } 12.3 \text{ ms, } 30 \text{ accepted intervals. } (203)$$

The corresponding scaled trajectory errors are approximately  $8.92 \times 10^{-5}$ ,  $5.93 \times 10^{-5}$ , and  $5.43 \times 10^{-5}$  for RBF, CGL, and LGL, respectively.

Unlike the inverter benchmarks, the event-free portions of the synchronous machine trajectory are highly smooth. High-order Chebyshev and Legendre polynomials can consequently approximate comparatively long temporal intervals with a small number of coefficients. The CGL and LGL formulations therefore satisfy the adaptive residual and off-grid audit conditions using substantially fewer accepted intervals than the RBF representation.

Radau and BDF are also particularly efficient for this benchmark. The synchronous-machine model is a relatively small and smooth stiff differential-algebraic system between discrete events, which corresponds closely to the class of problems for which mature implicit integrators are highly optimized.

Radau and BDF advance the physical states directly and employ efficient adaptive step-size selection, Newton iterations, and Jacobian reuse. BDF can additionally vary its integration order according to the local smoothness of the trajectory. For this problem, these algorithms therefore operate close to their most favorable numerical regime.

RAPTOR performs a different computation on each accepted interval. Rather than determining only the state at the next time point, it constructs a continuous local approximation

$$\hat{\mathbf{x}}(t) = \mathbf{x}_0 + \sum_{j=1}^p \mathbf{w}_j \phi_j(t),$$

where  $\phi_j(t)$  denotes a Gaussian temporal basis function. The coefficients must satisfy the governing equations at multiple collocation locations. Consequently, every interval requires basis and derivative evaluation, assembly of the physics residual and its Jacobian, solution of a coefficient-level nonlinear system, and an independent off-grid audit of the resulting continuous trajectory.

This richer interval representation introduces a fixed computational overhead. For the synchronous-machine trajectory, RAPTOR requires approximately 49–51 accepted temporal advances, whereas the polynomial representations can use longer intervals. The additional flexibility of the Gaussian representation therefore does not compensate for its interval-wise coefficient-solution cost in this particular model.

The synchronous-machine result consequently identifies a useful computational crossover. For small, smooth, stiff systems, established Radau and BDF integrators can remain more efficient because a continuous high-dimensional temporal representation is not required to resolve the local dynamics. The advantage of RAPTOR becomes stronger as the dynamics contain increasingly localized nonlinear behavior, interacting time scales, control actions, and event-induced transients, as illustrated by the EMT PLL experiment.

6) *Effect of exploiting model structure:* A separate implementation experiment was conducted to determine how much of the synchronous-machine runtime originates from the generic RAPTOR assembly procedure. After exact recovery of the algebraic variables, the smooth event-free dynamics admit an affine reduced representation. This allows the interval operators to be preassembled and factorized. ■

Applying this optimization consistently to the three RAPTOR bases gives first-use runtimes of approximately

11.90 ms (RBF), 12.19 ms (CGL), 12.34 ms (LGL).

These results show that a substantial fraction of the generic RBF cost in the synchronous-machine benchmark originates from repeated model assembly and coefficient-system construction rather than from the Gaussian basis functions themselves. Since this structure-aware experiment uses a different implementation strategy from the main adaptive benchmark, it is reported only as an implementation study and is not substituted for the main benchmark results.

7) *Discussion:* The experiments show that the most effective temporal representation depends on the dynamical structure of the problem.

For the RMS inverter, the Gaussian-RBF representation provides both a smaller error and a lower runtime than CGL and LGL under the same temporal approximation budget. For the more nonlinear and event-driven EMT PLL model, RAPTOR provides the lowest runtime among the methods satisfying the complete accuracy requirements.

The EMT synchronous-machine benchmark provides an important complementary result. Its low-dimensional, smooth, stiff dynamics strongly favor both polynomial collocation and mature implicit integrators such as Radau and BDF. In this regime, the additional computation required to construct and audit a continuous RBF approximation over every interval is not compensated by a reduction in the number of temporal advances.

The results therefore support a problem-dependent performance interpretation: RAPTOR is particularly attractive for nonlinear, multiscale, and event-driven transient simulations where localized temporal approximation is valuable, whereas classical implicit time integrators remain highly efficient for small and smoothly evolving stiff systems.

## REFERENCES



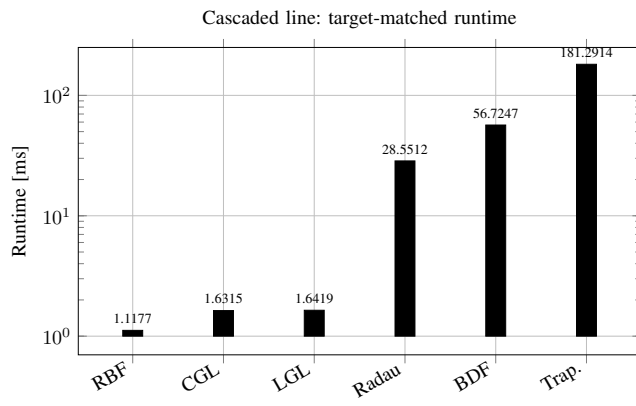


Fig. 6: Adaptive equal-size comparison for the EMT synchronous-machine benchmark. The smooth event-free machine dynamics favor the polynomial CGL and LGL representations, which satisfy the prescribed accuracy requirements using fewer accepted intervals than the Gaussian-RBF representation.